

STEFAN ZETZSCHE

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PROFILE

Applied Scientist at Amazon Web Services and former Meta intern, developing AI systems for code reasoning, reliable program synthesis, and formal verification in production-scale environments. Published at NeurIPS, PLDI, POPL, and ICML.

EMPLOYMENT

Amazon Web Services, Applied Scientist Dec 2022 - Present

- Use high-quality formal verification data to pre-train and fine-tune LLMs, enhancing their reasoning over low-level code and program behavior.
- Reduce LLM hallucinations in Amazon Bedrock and improve code generation in Kiro, Amazon's agentic IDE.
- Apply fully autonomous, agentic testing to detect correctness and security issues in software, uncovering multiple previously unknown Python standard library bugs.
- Contribute to Dafny's design and verification infrastructure and formally verify probabilistic algorithms in the Lean theorem prover for AWS Clean Rooms.

Meta, Software Engineer Intern Jul 2022 - Sep 2022

- Enhanced a static analyzer for the Hack programming language, improving code reliability in Meta's core infrastructure.

Amazon Web Services, Applied Scientist Intern Aug 2021 - Nov 2021

- Developed a formally verified quantum circuit optimizer in Dafny, enabling reliable quantum program compilation in collaboration with Braket, Amazon's quantum computing service.

HONORARY

University College London, Visiting Researcher Apr 2026 - Present

- Hosted by Prof. He Ye in the Software Systems Engineering Group.

EDUCATION

PhD Computer Science, University College London 2018 - 2023

- Conducted research on automata learning, Kleene algebra, and category theory, culminating in the thesis *Canonical Algebraic Generators in Automata Learning*.
- Funded by VeTTs and ERC grants; member of the Programming Principles, Logic, and Verification Group. Advised by Alexandra Silva and Matteo Sammartino.

MSc Mathematics, University of Hamburg 2016 - 2018

- Thesis on category theory (*Generalised Duality Theory for Monoidal Categories and Applications*), awarded top mark (1.0); graduated with highest distinction (overall grade 1.0).

BSc Mathematics, University of Hamburg 2014 - 2016

- Thesis on group and graph theory (*Isomorphism Classes of Vertex-Transitive Tournaments*), awarded top mark (1.0); minor in Computer Science.

PUBLICATIONS

Conferences

- [C1] **MiniF2F-Dafny: LLM-Guided Mathematical Theorem Proving via Auto-Active Verification** 2026
M. Bakšys, S. Zetzsche, O. Bouissou, S. B. Holden
International Conference on Machine Learning (ICML)
- [C2] **CLEVER: A Curated Benchmark for Formally Verified Code Generation** 2025
A. Thakur, J. Lee, G. Tsoukalas, M. Sistla, M. Zhao, S. Zetzsche, G. Durrett, Y. Yue, S. Chaudhuri
Conference on Neural Information Processing Systems (NeurIPS)

- [C3] **Verified Foundations for Differential Privacy** 2025
M. Medeiros, M. Naveed, T. Lepoint, T. Kahsai, T. Ravitch, [S. Zetsche](#), A. Joshi, J. Tassarotti,
A. Albarghouthi, J. Tristan
Distinguished Artifact Award
Conference on Programming Language Design and Implementation (PLDI)
- [C4] **Compiler Fuzzing in Continuous Integration: a Case Study on Dafny** 2025
K. Boonriong, [S. Zetsche](#), A. F. Donaldson
International Conference on Software Testing, Verification and Validation (ICST)
- [C5] **Well-Behaved (Co)algebraic Semantics of Regular Expressions in Dafny** 2024
[S. Zetsche](#), W. Róžowski
International Colloquium on Theoretical Aspects of Computing (ICTAC)
- [C6] **Generators and Bases for Monadic Closures** 2023
[S. Zetsche](#), A. Silva, M. Sammartino
Conference on Algebra and Coalgebra in Computer Science (CALCO)
- [C7] **Guarded Kleene Algebra with Tests: Automata Learning** 2022
[S. Zetsche](#), A. Silva, M. Sammartino
Conference on Mathematical Foundations of Programming Semantics (MFPS)
- [C8] **Canonical Automata via Distributive Law Homomorphisms** 2021
[S. Zetsche](#), G. v. Heerdt, M. Sammartino, A. Silva
Conference on Mathematical Foundations of Programming Semantics (MFPS)

Workshops

- [W1] **ATLAS: Automated Toolkit for Large-Scale Verified Code Synthesis** 2026
M. Bakšys, [S. Zetsche](#), O. Bouissou, S. Kong, R. Delmas
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)
- [W2] **MiniF2F-Dafny: LLM-Guided Mathematical Theorem Proving via Auto-Active Verification** 2026
M. Bakšys, [S. Zetsche](#), O. Bouissou
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)
- [W3] **DafnyPro: LLM-Assisted Automated Verification for Dafny Programs** 2026
D. Banerjee, O. Bouissou, [S. Zetsche](#)
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)
- [W4] **CLEVER: A Curated Benchmark for Formally Verified Code Generation** 2025
A. Thakur, J. Lee, G. Tsoukalas, M. Sistla, M. Zhao, [S. Zetsche](#), G. Durrett, Y. Yue, S. Chaudhuri
AI for Math Workshop at the International Conference on Machine Learning (ICML)
- [W5] **Well-Behaved (Co)algebraic Semantics of Regular Expressions in Dafny** 2025
[S. Zetsche](#), W. Róžowski
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)
- [W6] **Verifying the Fisher-Yates Shuffle Algorithm in Dafny** 2025
[S. Zetsche](#), T. Lepoint, J. Tristan, M. Mayer
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)
- [W7] **Dafny as Verification-Aware Intermediate Language for Code Generation** 2025
Y. C. Li, [S. Zetsche](#), S. Somayyajula
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)
- [W8] **Randomised Testing of the Dafny Compiler: Into the CI** 2025
K. Boonriong, A. F. Donaldson, [S. Zetsche](#)
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)
- [W9] **VMC: a Dafny Library for Verified Monte Carlo Algorithms** 2024
F. Zaiser, [S. Zetsche](#), J. Tristan
Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)

SUPERVISION

Interns at Amazon Web Services

- Jonas Bayer (PhD Student at the University of Cambridge) 2025

• Mantas Bakšys (PhD Student at the University of Cambridge)	2025
• Debangshu Banerjee (PhD Student at the University of Illinois Urbana-Champaign)	2025
• Yue Chen Li (Undergraduate Student at MIT)	2024
• Wojciech Różowski (PhD Student at the University College London)	2024
• Fabian Zaiser (PhD Student at the University of Oxford)	2023
• Yann Herklotz (PhD Student at the Imperial College London)	2022

TEACHING

Teaching Assistant at University College London

• Logic and Database Theory	2020
• Discrete Mathematics for Computer Scientists	2020
• Computability and Complexity	2020
• Theory of Computation	2019
• Principles of Programming	2019
• Discrete Mathematics for Computer Scientists	2018

Teaching Assistant at University of Hamburg

• Analysis I	2017
• Linear Algebra and Analytic Geometry II	2016
• Linear Algebra and Analytic Geometry I	2015

TALKS

[T1] COMP0010 Software Engineering module at the University College London	2025
[T2] Dafny Workshop at the Symposium on Principles of Programming Languages (POPL)	2025
[T3] International Colloquium on Theoretical Aspects of Computing (ICTAC)	2024
[T4] South of England Regional Programming Languages Seminar (SREPLS) at Jane Street	2024
[T5] Theoretical Computer Science Seminar Series at the University of Birmingham	2023
[T6] Oxford Advanced Seminar on Informatic Structures at the University of Oxford	2023
[T7] Programming Languages Discussion Group at Cornell University	2023
[T8] Lectures on Logic and its Mathematical Aspects Seminar at the University of Amsterdam	2023
[T9] Programming Principles, Logic, and Verification Seminar at the University College London	2023
[T10] Conference on Algebra and Coalgebra in Computer Science (CALCO)	2023
[T11] Conference on the Mathematical Foundations of Programming Semantics (MFPS)	2022
[T12] Programming Principles, Logic, and Verification Seminar at the University College London	2022
[T13] Lectures on Logic and its Mathematical Aspects Seminar at the University of Amsterdam	2022
[T14] Symposium on Compositional Structures (SYCO) at the Tallinn University of Technology	2021
[T15] Conference on the Mathematical Foundations of Programming Semantics (MFPS)	2021
[T16] Programming Principles, Logic, and Verification Seminar at the University College London	2021
[T17] Programming Principles, Logic, and Verification Reading Group at the University College London	2020
[T18] VeTSS Verified Software Workshop at the Isaac Newton Institute for Mathematical Sciences	2019
[T19] Scottish Programming Languages and Verification Summer School	2019
[T20] School of Computer Science at the University of Birmingham	2018

PROFESSIONAL SERVICES

Program Committee Chair

- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2026
- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2025
- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2024

Program and (Sub-)Review Committee Member

- International Conference on Machine Learning (ICML) 2025
- International Conference on Computer-Aided Verification (CAV) 2025
- Conference on Algebra and Coalgebra in Computer Science (CALCO) 2023
- International Symposium on Model Checking of Software (SPIN) 2023

Mentor

- Symposium on Principles of Programming Languages (POPL) 2021
- Conference on Systems, Programming, Languages and Applications: Software for Humanity (SPLASH) 2020

Organizer

- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2026
- South of England Regional Programming Languages Seminar (SREPLS) 2025
- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2025
- Dafny Workshop at the Symposium on Principles of Programming Languages (POPL) 2024

Artifact Evaluator

- International Conference on Computer-Aided Verification (CAV) 2025
- Conference on Programming Language Design and Implementation (PLDI) 2025
- International Conference on Functional Programming (ICFP) 2025
- Symposium on Principles of Programming Languages (POPL) 2025
- Conference on Systems, Programming, Languages and Applications: Software for Humanity (SPLASH) 2025
- International Conference on Functional Programming (ICFP) 2024
- Conference on Programming Language Design and Implementation (PLDI) 2024
- Symposium on Principles of Programming Languages (POPL) 2024
- International Conference on Computer-Aided Verification (CAV) 2023
- International Conference on Computer-Aided Verification (CAV) 2022